

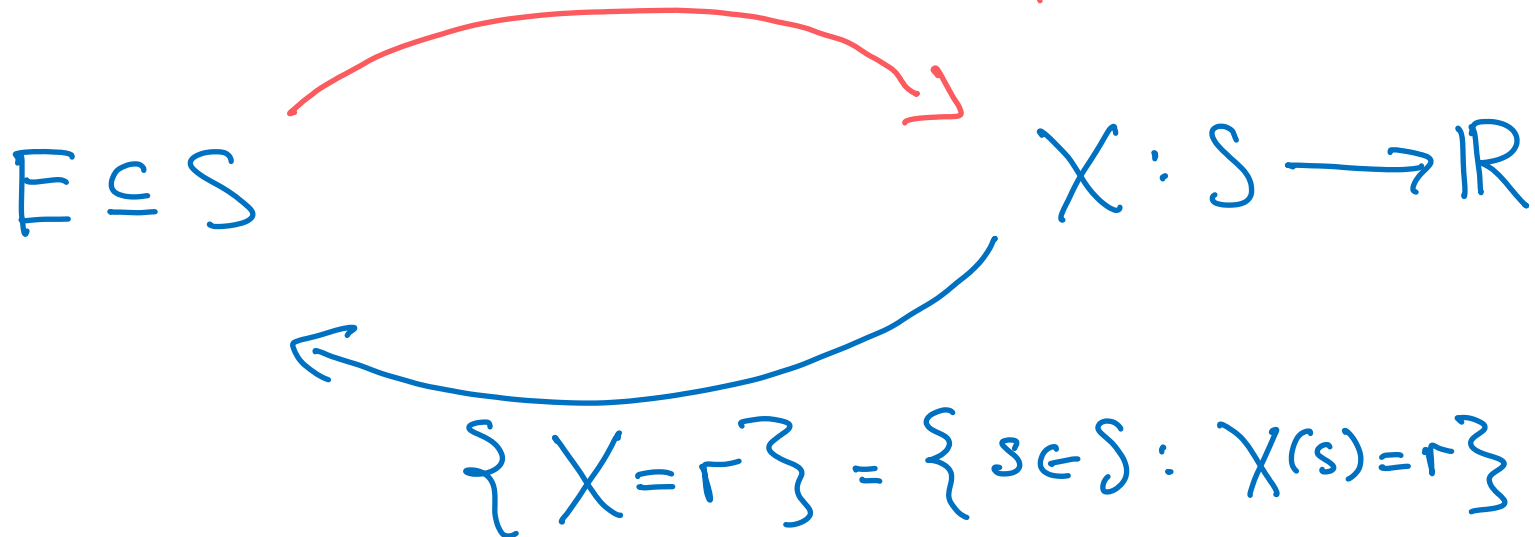
Lecture 25.

So far: Experiment, S , $p: S \rightarrow \mathbb{R}$

↑ Events $E \subseteq S$, $P(E)$, $P(E|F)$, independent.

↓ Random Variables $X: S \rightarrow \mathbb{R}$, $\mathbb{E}X$, $\mathbb{E}(X+Y) = \mathbb{E}X + \mathbb{E}Y$.

$$1_E(s) = \begin{cases} 1 & \text{if } s \in E \\ 0 & \text{if } s \notin E \end{cases} \quad \text{indicator random variable.}$$



Remark: $\mathbb{E} 1_E = 0 \cdot \cancel{P(1_E=0)} + 1 \cdot P(1_E=1) = P(E).$

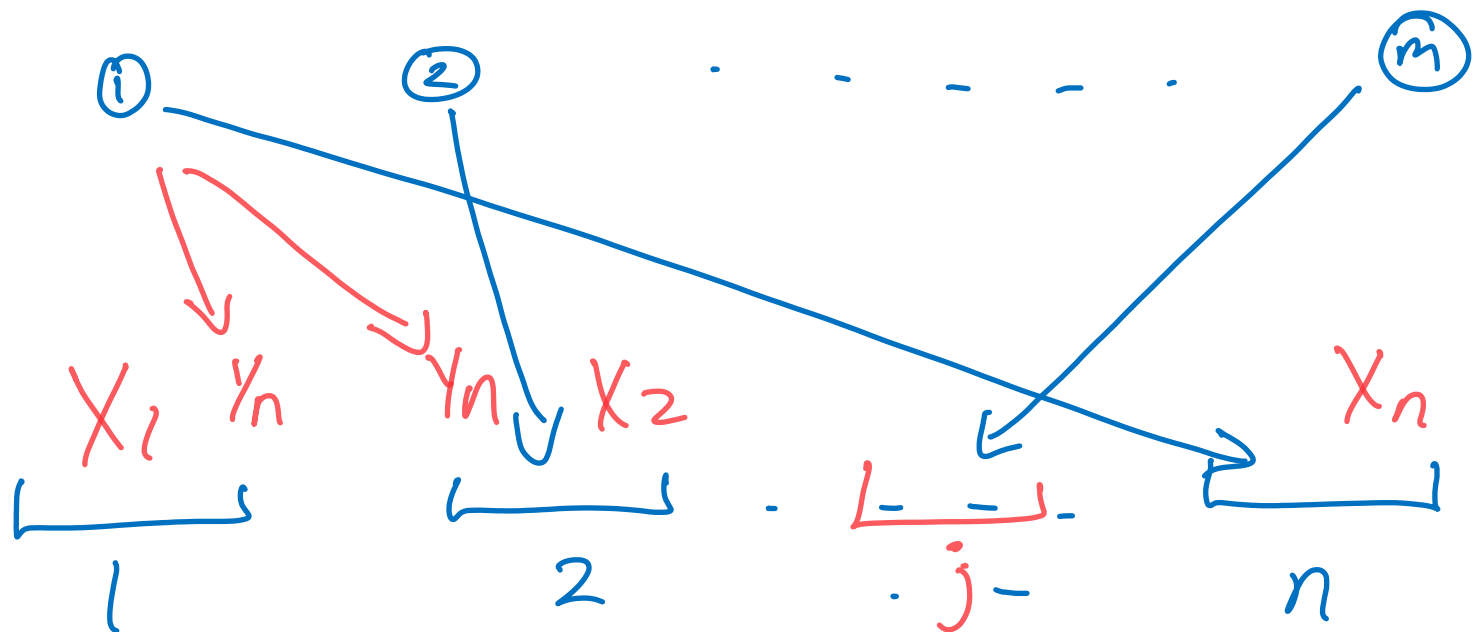
I Examples

II Probabilistic Method

III Markov's Ineq.

ex1: How many boxes of cereal do you need to
(Coupon Collector) by to collect all the power ranger?

Expt: Throw m balls into n labeled boxes
independently, uniformly at random.



(Q₁) How large should m be so that no box is likely to be empty?

(Q₂) If $m=n$, what is the expected number of empty bins? $\mathbb{E}X = ?$

(A₂) $S = \{ \overset{s}{(b_1, \dots, b_m)} : b_i \in [n] \}$ b_i : label of bin for ball i .

$X =$ # of empty bins.

idea: Define $X_j = \begin{cases} 1 & \text{if } \underline{\text{Bin } j \text{ is empty}} \\ 0 & \text{if not} \end{cases}$ $\left(\begin{matrix} \forall i \leq m \\ b_i \neq j \end{matrix} \right)$

Observe: $X = X_1 + X_2 + \dots + X_n$.

$$\left(\forall \underline{b} \in S \quad X(\underline{b}) = X_1(\underline{b}) + X_2(\underline{b}) + \dots + X_n(\underline{b}) \right).$$

Linearity: $\mathbb{E}X = \mathbb{E}X_1 + \mathbb{E}X_2 + \dots + \mathbb{E}X_n.$

$$\sim \frac{1}{e} + \dots + \frac{1}{e} = \underline{\underline{\frac{n}{e}}} \cdot \underline{\underline{37\% \text{ empty}}}.$$

Calculate: let $j \leq n.$

$$\mathbb{E}X_j = \overset{\text{indicator}}{P}(\text{bin } j \text{ is empty})$$

$$= P(\{b_1 \neq j\} \cap \{b_2 \neq j\} \cap \dots \cap \{b_m \neq j\})$$

$$\overset{\text{independent}}{=} P(\{b_1 \neq j\}) P(b_2 \neq j) \dots P(b_m \neq j)$$

$$\overset{\text{v.o.f.}}{=} \left(\frac{n-1}{n}\right) \left(\frac{n-1}{n}\right) \dots \left(\frac{n-1}{n}\right) \text{ } m \text{ times}$$

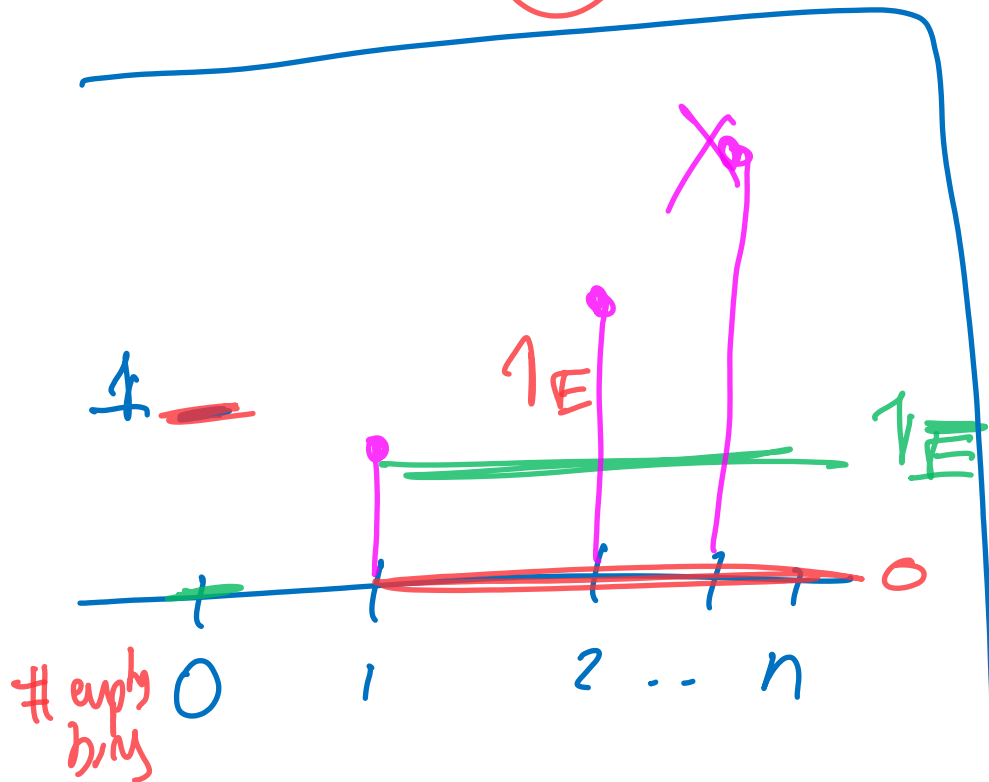
$$= \left(1 - \frac{1}{n}\right)^m \overset{m=n}{\sim} \frac{1}{e}.$$

(A₁) First Approach: Fix m, n . $S = \{(b_1, \dots, b_m) : b_i \in [n]\}$

Let $E = \{\text{No bin is empty}\} = \{X = 0\}$.
 $\bar{E} = \{\text{there is an empty bin}\}$

$$P(\bar{E}) = E 1_{\bar{E}}$$

Inequality $\leq E X$



Observation:

$\forall s: 1_{\bar{E}}(s) \leq X(s)$.

Fact: If $X(s) \leq Y(s) \forall s$,
then $E X \leq E Y$.

$$= \left(1 - \frac{1}{n}\right)^m + \dots + \left(1 - \frac{1}{n}\right)^m \quad n \text{ times.}$$

$$= \left[\left(1 - \frac{1}{n}\right)^n\right]^{m/n} \cdot n$$

$$\sim e^{-m/n} \cdot n.$$

$$P(E) \sim 1 - e^{-m/n} \quad \overset{m=2n \ln(n)}{\sim} 1 - \frac{1}{n^2} \cdot n = \frac{1}{n}$$

Recall: If $m = 2n \ln n$, $e^{-m/n} = e^{-2 \ln(n)}$

If $m > n \ln n$ then $P(E)$ is close to 1.

Can Show: If $m < n \ln n$ then $P(E) \sim 0$.

Will Show: If $m < n \ln(n)$ then $\mathbb{E} X \geq 1$.

Proof:

$$\boxed{\mathbb{E} X \sim e^{-m/n} \cdot n} \geq e^{-\frac{n \ln(n)}{n} \cdot n} = \frac{1}{n} \cdot n = 1 \quad \square$$

ex2: n students $\{1, \dots, n\}$ took the midterm.

Lost names.

Give exams back v.a.r., each student gets one exam.

(Q₁) Let $X = \#$ of students who get their exam.
What is $\mathbb{E}X$?

(A₁) $S = \{ (\pi_1, \dots, \pi_n) : \pi \text{ is a permutation of } [n] \}$.

Let $X_i = \begin{cases} 1 & \text{if } \pi_i = i \\ 0 & \text{if } \pi_i \neq i \end{cases}$. Observe: $X = X_1 + \dots + X_n$.

By linearity $\mathbb{E}X = \mathbb{E}X_1 + \mathbb{E}X_2 + \dots + \mathbb{E}X_n$.
 $= \frac{1}{n} + \dots + \frac{1}{n} = \underline{\underline{1}}.$

Calculate: If $i \leq n$, $\mathbb{E} X_i$

$$= P(\pi_i = i) = \frac{(n-1)!}{n!} = \underline{\underline{\frac{1}{n}}}.$$

(Q2) What is $P(X=0)$?

(A1) $P(X=0) = P(\pi \text{ is a derangement})$

$$= \frac{|D_n|}{n!} \sim \frac{1}{e}.$$

Summary: Creativity is in defining the right events and random variables to decompose complicated phenomena into simple ones.

- $P(E_1 \cap E_2) = P(E_1)P(E_2)$ independence
- $\mathbb{E} X_1 + X_2 = \mathbb{E} X_1 + \mathbb{E} X_2$ always holds.

II Probabilistic Method.

Ramsey's Theorem: $\forall k \geq 3 \quad \exists N$ s.t.

every graph with at least N vertices
contains a k -clique or a k -independent set
as an induced subgraph.

< Proved: when $k=3$, $N=6$ is enough >.

(Q) For a given k , what is the largest
graph that has neither?
 Let $R(k)$ be its # of vertices

(A) For $k=3$, $R(k)=5$.

$$k=4, R(k)=17.$$

$$k=5 \quad \underline{\text{unknown}}.$$

$$\rightarrow \text{So: } R(k) \gg \lfloor 2^{k/2} \rfloor.$$

Theorem (Erdős): $\forall k \geq 3$ there is a graph G on $N = \lfloor 2^{k/2} \rfloor$ vertices s.t. G has no k -clique and no k -independent set.

Uniform set

Proof: Let $k \geq 3$ and let $N = \lfloor 2^{k/2} \rfloor$.

Let G be a uniformly random graph on

$\{1, \dots, N\}$.

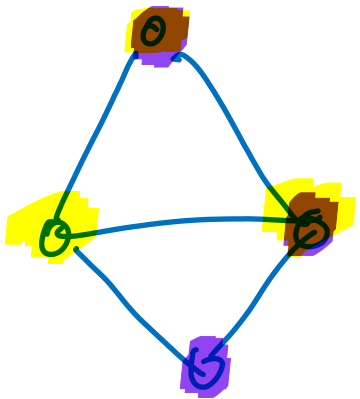
$$S = \{ G : G \text{ is a graph on } [N] \}$$

→ Can generate by flipping a fair coin for each pair $x, y \in [N]$.

$$|S| = 2^{\binom{N}{2}}.$$

Call $T \subseteq [N]$, $|T|=k$ uniform if the subgraph of G induced by T is either a k -clique or a k -independent set.

$k=3$



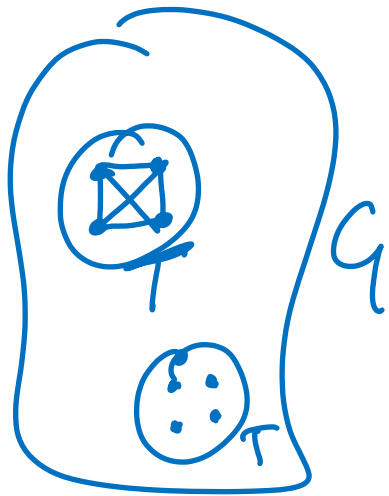
Let $X = \#$ of uniform subsets of $[N]$

Goal: $\mathbb{E} X < 1$ for $N \geq \lceil 2^{k/2} \rceil$.

Observe that this implies $P(X=0) > 0$:

If $P(X=0)=0$ then $\mathbb{E} X = \sum_{r=1}^{\infty} r P(X=r) \geq 1$.

idea: For every $T \subseteq [N]$ with $|T|=k$
 let $X_T = \begin{cases} 1 & \text{if } T \text{ is uniform} \\ 0 & \text{otherwise.} \end{cases}$



Observe: $X = \sum_{\substack{|T|=k \\ T \subseteq [N]}} X_T.$

By Linearity: $\mathbb{E} X = \sum_{\substack{|T|=k \\ T \subseteq [N]}} \mathbb{E} X_T.$

Calculate: Fix $T \subseteq [N]$.

$$\mathbb{E} X_T = \mathbb{P}(T \text{ is uniform})$$

$$= \frac{2}{2^{\binom{k}{2}}} \begin{array}{l} \text{\# of uniform graphs on } T \\ \text{\# of all graphs on } T. \end{array}$$

$$\therefore \mathbb{E} X = \sum_{\substack{|T|=k \\ T \subseteq [N]}} \frac{2}{2^{\binom{k}{2}}} = \binom{N}{k} \frac{2}{2^{\binom{k}{2}}}$$

$$= \frac{N(N-1) \dots (N-k+1)}{k!} \frac{2}{2^{\binom{k}{2}}}$$

$$\leq \frac{N^k}{k!} 2^{1-\binom{k}{2}}$$

$$\leq \frac{\left(2^{\frac{k}{2}}\right)^k 2^{1 - \frac{k(k-1)}{2}}}{k!}$$

$N = \lfloor 2^{k/2} \rfloor$

$$= \frac{2^{\cancel{\frac{k^2}{2}} + 1 - \cancel{\frac{k^2}{2}} + \frac{k}{2}}}{k!} \leq \frac{2^{\frac{k}{2} + 1}}{k!}$$

$$< 1. \quad \square$$

Not known how to construct such Q .

